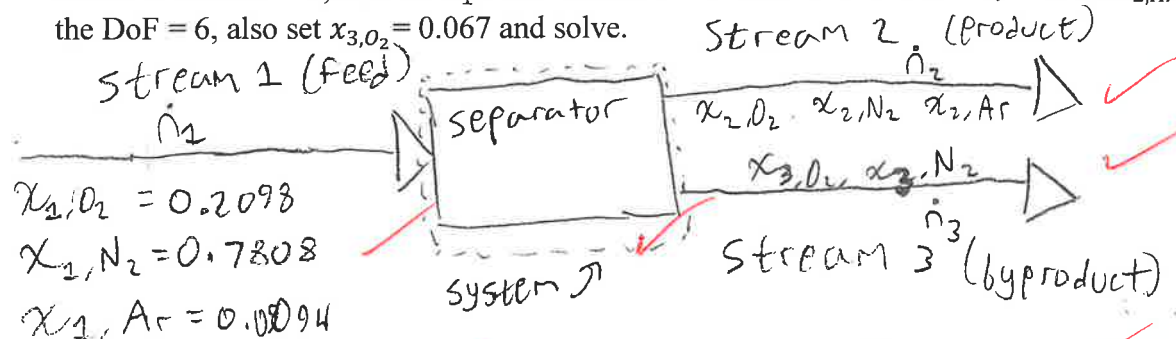


To Receive Full Credit, You Must Show All Your Work

The Linde-Frankl separation process uses a feed of atmospheric air (Stream 1) which has a composition of 20.98 mol% oxygen (O_2), 78.08 mol% nitrogen (N_2), and 0.94 mol% argon (Ar) to produce a product (Stream 2) enriched in oxygen and a byproduct (Stream 3) which is enriched in nitrogen and contains no argon.

- (a) Draw and label a block flow diagram. Define a system. For the 3 streams, use $\dot{n}_i$ for total molar flow rates and x_{i,O_2} , x_{i,N_2} , and $x_{i,Ar}$ for mole fractions for oxygen, nitrogen, and argon, respectively. Use $i = 1, 2$, & 3.
- (b) Do a DoF analysis using p. 105 of Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DoF.
- (c) If the DoF < 0, stop here. The problem is overspecified. If DoF = 0, solve the problem. If DoF = 1, set $\dot{n}_2 = 100$ kmol/hr and solve. If DoF = 2, also set $x_{2,O_2} = 0.9$ and solve. If the DoF = 3, also set $x_{3,N_2} = 0.953$ and solve. If the DoF = 4, also set $\dot{n}_1 = 224$ kmol/hr and solve. If the DoF = 5, also set $x_{2,Ar} = 0.091$ and solve. If the DoF = 6, also set $x_{3,O_2} = 0.067$ and solve.



DoF $4a = (3+3+2)$, $4b = 0$

$5a = 0$, $5b = (2+0+0)$, $5c = 0$, $5d = 3$

$4a + 4b - 5(a+b+c+d) = 8 - 5 = 3$, so $\dot{n}_2 = 100$ kmol/hr, $x_{2,O_2} = 0.9$

species balance	x_1	$\dot{n}_1$	x_2	$\dot{n}_2$	x_3	$\dot{n}_3$
O_2	0.2098	46.995	0.9	90	0.047	43.0
N_2	0.7808	174.89	0.078	7.89	0.953	79.6
Ar	0.0094	2.106	0.021	2.1	0	0
Tot	1	224	1	100	1	122.6

$\dot{n}_1 \cdot 0.2098 = 100 \cdot 0.9 + 0.047 \cdot \dot{n}_3$

$\dot{n}_1 \cdot 0.7808 = 100 \cdot x_{2,N_2} + 0.953 \cdot \dot{n}_3$

$\dot{n}_1 \cdot 0.0094 = 100 \cdot x_{2,Ar} + 0$

$224 \cdot 0.0094 = x_{2,Ar} \cdot 100$
 $x_{2,Ar} = 0.021$

DoF $4a = 8$, $4b = 0$

$5a = 1$, $5b = (2+0+1)$, $5c = 0$, $5d = 3$

$8 - 7 = 1$, so $\dot{n}_1 = 224$ kmol/hr
 solution incorrect.